

AKASH P

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github.com/akashp2002

Summary

MCA graduate with a strong technical foundation in full-stack software development and Artificial Intelligence. Proficient in MERN stack and Java, complemented by practical skills in machine learning, NLP, and RAG pipelines. Experienced in building end-to-end applications and dedicated to applying foundational AI concepts to solve complex problems within collaborative, high-impact engineering environments.

Education

Tkm College of Engineering Master of Computer Application	2024 – 2026
Morazha Co-Operative Arts and Science College Bachelor of Computer Application	2019 – 2022

Projects

Cloudflare Incident RAG System	https://github.com/akashp2002/CloudflareAssistant
- Developed a full-stack RAG system using Python, FastAPI, React, LlamaIndex, Qdrant, Groq, and Docker to answer questions from Cloudflare incident postmortems. - Implemented hybrid retrieval by combining dense vector search and BM25 lexical search using Reciprocal Rank Fusion (RRF), followed by cross-encoder reranking for improved retrieval relevance. - Built grounded answer generation with source citations and explicit refusal for insufficient context, and evaluated RAG quality using Ragas. - Added end-to-end pipeline observability using LangSmith and containerized the FastAPI, React, and Qdrant services using Docker Compose.	
DeliverEase Web Application	https://github.com/akashp2002/DeliverEase
- Developed a Scheduled Delivery Management System using the MERN stack (MongoDB, Express.js, React.js, Node.js). - Implemented route optimization using a Distance Matrix and Nearest Neighbor algorithm to generate efficient delivery sequences. - Integrated interactive map visualization using Leaflet and OpenStreetMap for route display. - Added real-time GPS tracking and turn-by-turn voice navigation to guide delivery agents.	

Skills

Frontend: React js, HTML, CSS
BackEnd: Node js, Express js, FastAPI
Programming Languages: Java
Database: MongoDB, MySQL, Qdrant (Vector DB)
AI/GenAI: RAG Pipelines, RNNs, LlamaIndex, Transformers
Tools: Docker, Git, LangSmith, Ragas

Certificates

Data Structures and Algorithms Design – NPTEL–IIT Kanpur	2025
MERN Stack Development	2022–2023
Deep Learning – DeepLearning.AI (Coursera)	2026